CM617: SUSTAINABILITY PRINCIPLES AND PRACTICES
IIT BOMBAY

UNWRAP:

Unpacking Whether Replacement Alternatives Perform

Does sustainable packaging actually reduce carbon?
A paired meta-analysis of packaging life cycle assessments
under Indian collection rates and an honest functional unit

Course Project Report

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UNWRAP: Does replacing plastic packaging actually reduce carbon?

A paired meta-analysis of 46 published LCA comparisons, with an Indian collection-rate test and an honest functional unit

1. Introduction

Replacing plastic with glass, aluminium, cartons or refills is widely assumed to cut carbon, but life cycle assessments (LCAs) disagree. A review of 53 LCAs found plastics were often not the worst option and that end-of-life assumptions drove the result (Dolci et al., 2025). Three gaps remain. (i) Studies are rarely pooled *within* a study, where both packs share one method; cross-study averages mix method noise into the comparison (Bher and Auras, 2024). (ii) Results assume European or US waste systems, while India collects 85% of aluminium cans, 70% of PET and 60% of glass (TERI, 2022) and burns much uncollected waste (Cottom et al., 2024). (iii) Every LCA divides impact by product *filled*, ignoring product left stuck in the pack. We test three hypotheses: **H1** alternatives have lower GWP than plastic; **H2** any advantage shrinks under Indian collection rates; **H3** scoring packs per unit of product *used* changes their ranking.

2. Methods

Data collection. We searched Scopus, ScienceDirect, Google Scholar and grey literature for LCAs reporting 100-year GWP for a conventional and an alternative FMCG pack under one method. Of 90 records, 52 were assessed and 24 reports (23 studies) gave 46 extractable pairs (PRISMA flow, Fig. S1). Each pair was coded for material, funder, system boundary, functional unit, region, data source and whether end-of-life rates were stated. Eighteen values from an industry-commissioned compilation (Wohner and Tacker, 2020) were checked against its tables. The **core pool** holds plastic-baseline pairs that switch format for the same product (31 pairs, 16 studies); seven weak pairs (likely unit errors, different products, out of scope) enter sensitivity runs only.

Stage 1, meta-analysis. The effect size is $y = \ln(GWP_{alt}/GWP_{conv})$, so a ratio below 1 favours the alternative. Pairs were averaged within studies and studies pooled with equal weight (within-study variances are not reported), with study-level bootstrap 95% intervals (20,000 resamples). Method choices were tested on class-adjusted residuals; robustness by twelve sensitivity subsets and leave-one-out; funding bias by funder subsets.

Stage 2, India model. A Monte Carlo model (20,000 draws) computes GWP per litre for a 500 mL can, 600 mL PET and 650 mL glass bottle from TERI masses as production + conversion + transport − collection × yield × recycling credit + burning of uncollected PET, under European and Indian collection rates. c* is the can collection rate at which the can equals PET.

Stage 3, honest unit. For four shampoo packs we compare packaging GWP per kg filled (G/F) with GWP per kg *used*, (G + PF) / [F(1−r)(1−u)] − P, where r is measured residue (pump 7.1%, pouch 2.7%, canister 3.2%; sachet 3% to 11% from portion packs), u is never-used product and P the product footprint. Uncertain inputs were varied in Monte Carlo (Table S6).

3. Results

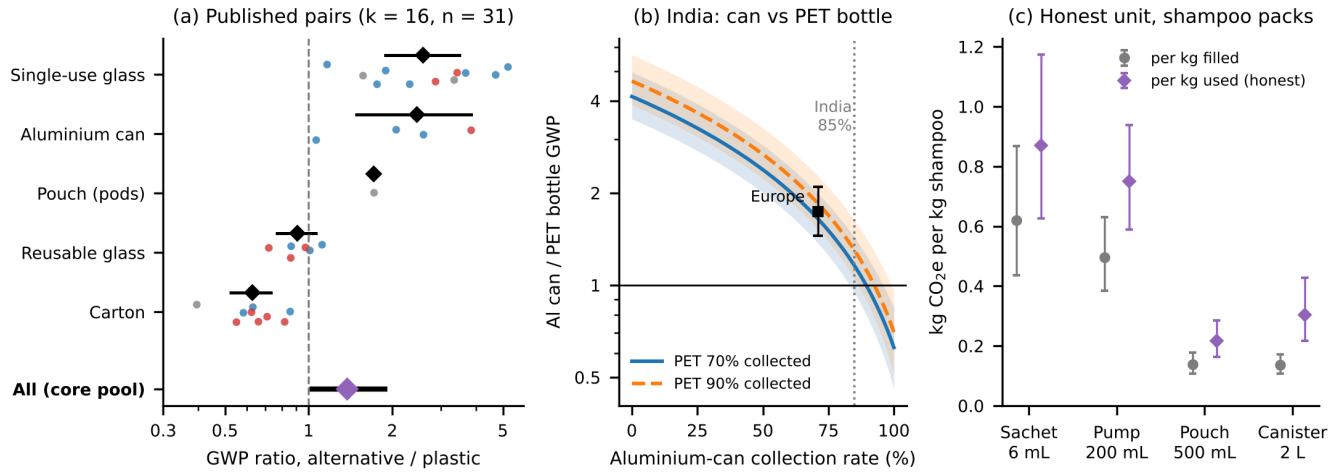


Fig. 1. (a) Published GWP ratios by alternative material (dots; blue academic, red industry-linked, grey other) with pooled estimates and bootstrap 95% intervals (diamonds). (b) Modelled aluminium can / PET ratio against can collection under Indian conditions (bands 5th to 95th percentile); square = European rates. (c) Shampoo-pack GWP per kg filled vs per kg actually used, India (bars 5th to 95th percentile).

H1, not supported. On average the alternative had 1.38 times the GWP of the plastic pack and was better in only 13 of 31 pairs (Table 1, Fig. 1a). The answer depends on the material: cartons won every pair (0.63x), reusable glass was at parity (0.91x), while single-use glass (2.58x) and aluminium (2.45x) never won. Leave-one-out estimates stayed between 1.26 and 1.45. The one method choice with a clear effect was data source: values passed through the industry compilation were about 26% more favourable to alternatives after adjusting for material (shift 0.74, 0.57 to 0.96). In 4 of 5 industry-linked studies the sponsor's own material ranked best.

Table 1. Pooled GWP ratio (alternative / plastic) by material and key robustness checks.

Group	Studies	Pairs	Ratio	95% interval	Alternative better
Single-use glass	9	11	2.58	1.89 to 3.49	0 of 11
Aluminium can	3	4	2.45	1.48 to 3.85	0 of 4
Reusable glass (5 to 8 uses)	4	6	0.91	0.77 to 1.06	4 of 6
Beverage carton	7	9	0.63	0.52 to 0.73	9 of 9
All, core pool (incl. 1 pouch pair)	**16**	**31**	**1.38**	**1.03 to 1.88**	**13 of 31**
Values from original sources only	8	13	1.82	1.16 to 2.83	4 of 13
Values from industry compilation only	8	18	1.04	0.79 to 1.38	9 of 18
Academic-funded studies only	8	16	1.54	1.02 to 2.42	4 of 16
Core pool without single-use glass	13	20	0.96	0.71 to 1.35	13 of 20

Bootstrap intervals over studies. Full sensitivity and leave-one-out results in Fig. S3; moderator tests in Table S2.

H2, partly supported. Under European rates the can emits 1.73x a PET bottle (1.45 to 2.09) and glass 5.3x. Under Indian rates the can's penalty falls to 1.16x (0.94 to 1.43), and the can beats PET in 13% of draws. Parity needs c* = 89% can collection (83% to 96%) against the reported 85% (Fig. 1b). Glass stays about 5x PET at any collection rate.

H3, supported. Counting stranded product raised every pack's footprint by 38% to 121% (Fig. 1c). Per kg filled, the 2 L canister tied with the 500 mL refill pouch (0.98x); per kg used it lost (1.40x, 1.03 to 1.93) and the pouch was best in 96% of draws. The sachet's penalty against a 200 mL pump bottle, which strands 7% of its shampoo, shrank from 1.25x to 1.16x but did not disappear.

4. Discussion

What works and why. Replacing plastic does not cut carbon in general. Cartons work because they are light and mostly paper. Reusable glass works only after several trips, consistent with Dolci et al. (2025), who place break-even near 10 reuses. Single-use glass fails because it is about 20 times heavier per litre. Aluminium is the most context-sensitive option: India's high can collection nearly closes the gap with PET, but parity needs almost every can to be collected, so the advantage is fragile. **Evidence quality.** Sponsor-favouring results and the more favourable values in an industry compilation argue for independent comparisons with stated end-of-life assumptions. **Functional unit.** The honest unit shows that the choice of unit can decide which pack wins. Packs that empty well (pouches, right-sized packs) beat simply larger ones, and a sachet ban helps only if users switch to refills rather than small pump bottles. **Limitations.** Only 24 of 44 eligible reports had open numbers, below our 60-pair target; studies are equally weighted because variances are unreported, so no funnel test was possible; no paired sachet or mono-material LCA exists, so Stage 3 is a model built on measured residues and assumed pack masses; only GWP was assessed. **Conclusion.** The carbon benefit of replacement packaging depends on the format, the local waste system and the functional unit. Packaging claims should be judged per unit of product actually used, under local collection rates, and on independent evidence.

Supplementary information. UNWRAP_supplementary.pdf (full methods, equations, PRISMA flow, full forest plot, sensitivity and moderator tables, model inputs, study characteristics); UNWRAP_data_and_results.xlsx (all 46 pairs with sources and coding, live formulas); analysis.py, build_workbook.py, paired_comparisons.csv, prisma_log.csv (fully reproducible, fixed seeds).

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All 23 included studies are listed in Table S5; full references in the Supplementary information.